May 1994

Pattern Recognition Letters

ELSEVIER

Pattern Recognition Letters 15 (1994) 507–517

Techniques for speeding up high-quality perspective Maximum Intensity Projection

K.J. Zuiderveld *, A.H.J. Koning, M.A. Viergever

Computer Vision Research Group, Utrecht University, Heidelberglaan 100, 3584 CX Utrecht, Netherlands

Received 29 July 1993; revised 18 October 1993

Abstract

This paper presents three classes of speed-up techniques for calculating high-quality perspective Maximum Intensity Projection (MIP) images from Magnetic Resonance Angiography (MRA) data. The method of choice is a combination of Ray Acceleration by Distance Coding (RADC) with Closest Vessel Projection (CVP); this technique combines a five- to ten-fold speed improvement with improved visualization of blood vessels.

Key words: Volume visualization; Speed-up techniques; Trilinear interpolation; Maximum intensity projection; Closest vessel projection; MIP; CVP; RADC

1. Introduction

Magnetic Resonance Angiography is a relatively novel acquisition technique that allows for visualization of blood vessels without the need for X-rays. Fig. 1 contains parts of two slices that were obtained using different MRA techniques; blood vessels in these images can be recognized by their high pixel intensity. Since an MRA acquisition results in a 3D dataset that can be viewed from any angle, it offers great promise for the display of vascular morphology and pathology.

Display of 3D MR angiograms poses a non-trivial problem; the volume data is often too noisy to calculate satisfactory 3D images using traditional 3D reconstructions based on extraction and shadowing of surfaces (Zuiderveld and Viergever, 1992). Therefore, in a clinical setting, MIP is usually the method of choice; with this ray casting technique, the maximum grey-value encountered along each ray is assigned to the corresponding pixel in the output image.

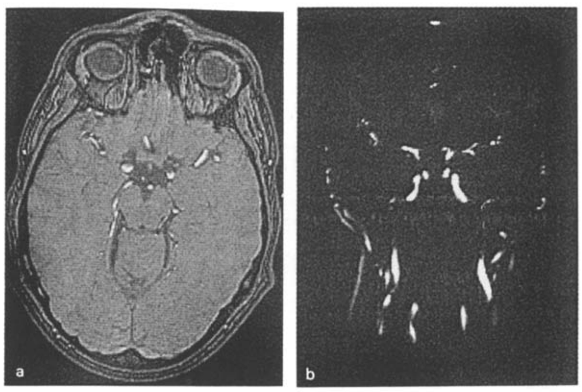

Fig. 1. Examples of MRA data. (a) Detail of a transversal slice from a 3D time-of-flight MRA acquisition. (b) Detail of a coronal (frontal) slice from a 3D phase contrast angio measurement from the vessels in head and neck, more specific the circle of Willis and the carotids.

While 3D surface display allows for adaptive ray

* Corresponding author. Email: karel@cv.ruu.nl

SSDI 0167-8655(93)E0097-8

termination (termination of ray casting as soon as the ray hits an opaque surface (Levoy, 1990), MIP requires the evaluation of all sample points along each ray. Consequently, a major component of the computational costs associated with MIP is the calculation of sample points along the rays, especially when perspective projection is used.

As any ray casting method, MIP also requires a geometrical transformation between object and screen space; this implies resampling (interpolation) of the volume data. Resampling is usually done by a zeroth-order (nearest neighbour interpolation) or a first-order interpolation algorithm (e.g., trilinear interpolation); an example of these methods can be seen in Fig. 2.

While higher-order interpolation algorithms are computationally more expensive than simple nearest neighbour interpolation, the resulting images have superior quality, since they do not contain the disturbing aliasing artifacts ("jaggies") that are typical for nearest neighbour interpolation.

In clinical applications, the overall speed as well as the perceived image quality is important; unfortunately, a tradeoff between these two factors must be made. Since patient throughput is often a dominant factor, physicians are usually forced to choose speed over quality. Instead of high-quality MIP reconstructions (using trilinear interpolation and perspective projection), MIP using nearest neighbour interpolation and parallel projection tends to be the method of choice.

This paper describes three classes of techniques that significantly speed-up high-quality MIP reconstructions, which is of great interest for clinical application of MIP. We first describe the Interval Analysis (IA) technique that uses precalculated local maxima to reduce the number of trilinear interpolations. Next, we describe a technique that builds upon IA and adds threshold-based interpolation, a technique where trilinear interpolation is performed only in case local maxima are higher than a volume-dependent threshold. We then explain the RADC technique where a 3D distance transform is used to skip non-interesting parts of the volume. The paper is concluded with a comparison of benefits and disadvantages of the methods described.

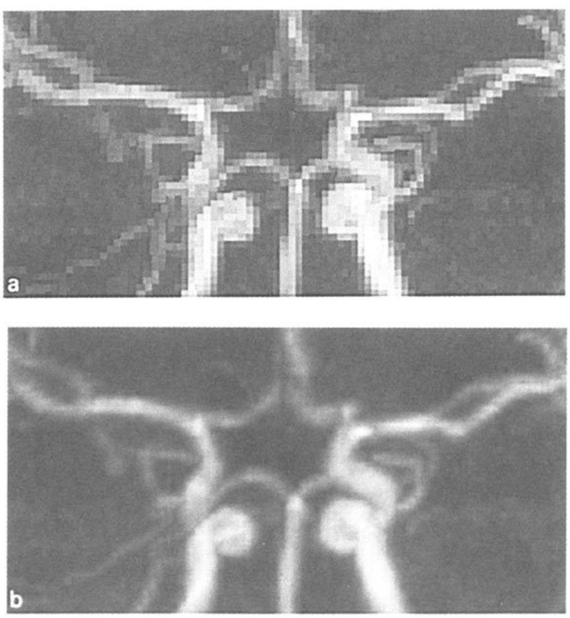

Fig. 2. Examples of MIP images. (a) MIP using nearest neighbour interpolation of grey-values along the ray. (b) MIP using trilinear interpolation of grey-values. This is a zoomed image that shows the circle of Willis.

2. Interval analysis

2.1. Basic principle

With trilinear interpolation, the grey-value of a ray sample is calculated as follows (see also Fig. 3):

$$\begin{aligned} G(x,y,z) = {} & (1-x)(1-y)(1-z)G_{000} \\ & +x(1-y)(1-z)G_{001} \\ & +(1-x)y(1-z)G_{010} \\ & +xy(1-z)G_{011} \\ & +(1-x)(1-y)zG_{100} \\ & +x(1-y)zG_{101} \\ & +(1-x)yzG_{110} \\ & +xyzG_{111} \end{aligned} \tag{1}$$

where x, y, and z represent the fractional part of the sample point coordinates, G_{ijk} the grey-values of the surrounding voxels and $G(x, y, z)$ the interpolated grey-value. Trilinear interpolation is expensive; an

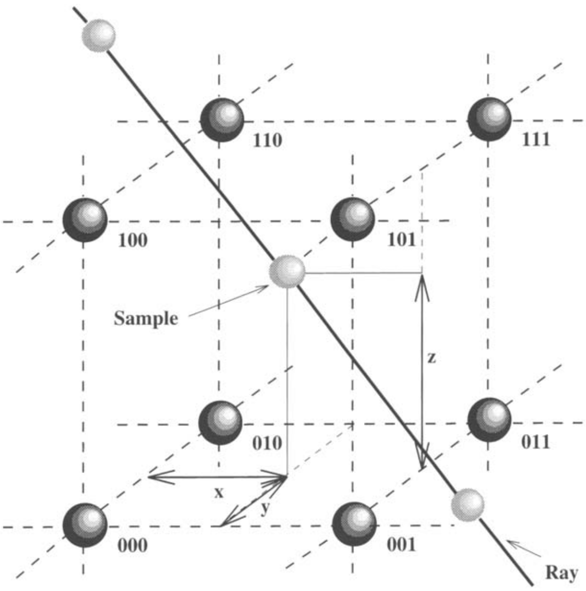

Fig. 3. Grid positions and ray sample positions used for trilinear interpolation.

optimized implementation typically requires fourteen multiplications and seven additions, while the fractional coordinates also have to be calculated.

As can be seen from Eq. (1), the interpolated grey-value $G(x, y, z)$ cannot be smaller than the minimum grey-value of the eight voxels that are involved in the interpolation; likewise, $G(x, y, z)$ cannot exceed the maximum of these voxels.

This observation has lead to a strategy called Interval Analysis. With this method, we calculate during a preprocessing step local maxima of the $3\times3\times3$ neighbourhood associated with each voxel in the original dataset and store the resulting values in a new dataset, which is denoted as the *maximum volume*. Likewise, one can also calculate a *minimum volume* containing the local minima of the MRA voxels. Fig. 4 contains a slice of the original MRA dataset as well as the corresponding slice from the minimum and maximum volumes.

MIP using Interval Analysis uses two datasets as input, namely the original MRA data and the corresponding maximum volume. For each sample step during the maximum intensity calculation, the nearest neighbour value associated with the maximum volume (the upper bound) is compared against the maximum value encountered so far. If the upper bound associated with the sample point has a lower value, trilinear interpolation will not result in a new maximum and can therefore be omitted.

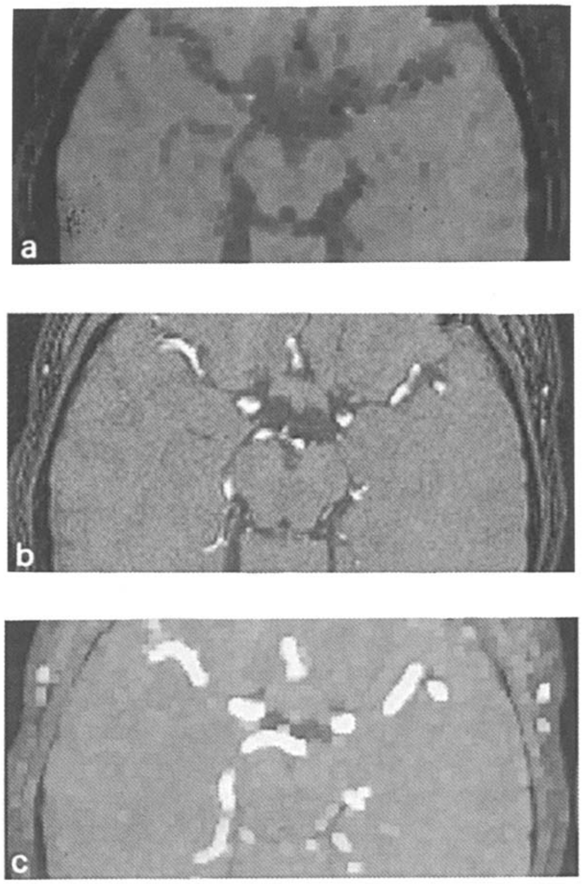

Fig. 4. Local extrema in a $3\times3\times3$ neighbourhood. (a) Calculated local minima. (b) Original MRA data. (c) Calculated local maxima. Note the differences in image intensities; while the blood vessels are clearly visible in the maximum image, they are not visible in the minimum image.

2.2. Results of standard interval analysis

We evaluated our methods using three MRA datasets. Two datasets were 3D phase-contrast MRA datasets (CA-1) and CA-2), while the third one (TA) is a time-of-flight MRA dataset (see Huston and Ehman (1993) for information on the relative merits of each technique). Each dataset has a different dimension; CA-1 has a dimension of $256\times256\times48$, CA-2 is $110\times200\times128$, and the TA dataset has a size of $174\times240\times122$. The latter two datasets were obtained by removing uninteresting regions from larger datasets.

The performance of Interval Analysis is summarized in Table 1. These measurements indicate that Interval Analysis reduces the number of trilinear interpolations by approximately 30 to 50%, but due to the overhead associated with Interval Analysis and ray casting, the actual improvement with respect to execution time is less.

These moderate improvements are caused by the signal characteristics of MRA datasets, which contain a significant amount of noise. Due to this noise, local maxima in the background areas tend to have high values, which results in trilinear interpolations as long as the ray does not hit a high-intensity voxel containing a blood vessel. Only when a sample located in a blood vessel is encountered, the grey-value associated with the ray becomes high enough to cause interpolations to be skipped. However, since only a small fraction (typically less than 2%) of the dataset contains blood vessel structures, many samples are evaluated using trilinear interpolation.

2.3. Lower-bound interval analysis

The results of Interval Analysis can be significantly improved if a lower bound on the MIP ray value is calculated prior to the actual MIP calculation. Fig. 5 illustrates two different implementations of this approach.

The first approach for calculating a lower-bound image relies on the use of a minimum volume that contains local minima calculated for a $3\times3\times3$ neighbourhood (see Fig. 4a for an example). A lower bound for trilinear MIP can be obtained by performing a nearest neighbour MIP of the minimum volume; the lower bound is then used as the initial grey-

Table 1
Reduction of the number of trilinear interpolations using standard Interval Analysis

Dataset	Fraction of interpolations	Improvement in execution speed
CA-1	67.9%	1.19
CA-2	51.9%	1.39
TA	48.4%	1.50

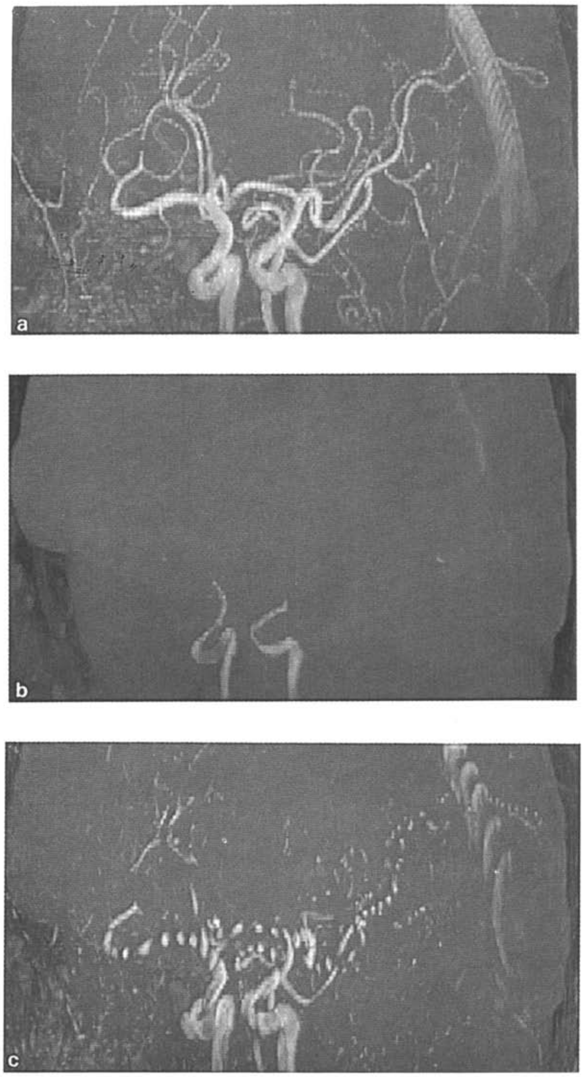

Fig. 5. Estimation of lower-bound value for MIP. (a) Standard trilinear MIP (sampling rate 2.0).)b) Lower-bound image calculated using nearest neighbour MIP of a volume dataset containing local minima. (c) Lower-bound image calculated using trilinear MIP with a sampling rate of 0.2

value of the ray when the actual trilinear MIP is performed.

Fig. 5b contains a nearest neighbour MIP of a minimum volume. Compared with the non-accelerated trilinear MIP depicted in Fig. 5a, only a small fraction of the vessels can be recognized. The major part of the lower-bound image contains background noise which is darker compared to that of the trilinear MIP.

Initial experiments with this method indicated that the number of trilinear interpolations was only marginally reduced, while execution times increased due to the added overhead. Since storage of the minimum volume also requires a considerable amount of memory, determination of a lower bound using a minimum volume should not be considered a viable option.

An alternative approach for obtaining a lower-bound estimate is to perform a trilinear MIP of the original dataset using a very low sampling rate; an example of a resulting lower-bound image can be found in Fig. 5c. One can clearly see the improvement with respect to the lower-bound estimation obtained by the minimum volume; since an additional dataset is also avoided, undersampled MIP is clearly the better method for obtaining a lower-bound image.

A potential disadvantage of the undersampled MIP is the additional overhead. With very low sampling rates (0.05 and lower), the overhead is relatively minor, but the obtained lower bound will have small values since only a few samples are evaluated along each ray. With higher sampling rates (0.25 and higher), the added overhead associated with ray casting and trilinear interpolations becomes prohibitive.

2.4. *Results of lower-bound interval analysis*

Fig. 6 summarizes some experiments that were performed to find an optimum value for the sampling rate associated with the undersampled MIP. The three datasets gave varying results.

For dataset CA-1, the number of trilinear interpolations is only marginally reduced; the obtained reduction is not large enough to justify the lower-bound calculation, especially when moderate undersampling is used. With very low sampling rates, the additional overhead is only slightly larger than the gain.

The two other datasets clearly benefit from the first pass. With dataset CA-2, the number of trilinear interpolations decreases with 15 to 30%, depending on the sampling rate. The TA dataset even shows an improvement of 35%, virtually independent of the sampling rate.

The differences in behaviour with respect to the lower-bound MIP are caused by the individual characteristics of the datasets. The TA dataset is a time-of-flight MRA acquisition, which besides the blood vessels also contains soft tissue; the 35% reduction can be contributed to the elimination of interpolations in the low intensity regions of the volume (background and bony tissue). In contrast, 3D phase contrast MRA only visualizes blood vessels; the reduction of the number of trilinear interpolations strongly depends on the fraction of voxels containing high intensities. Since the number of high intensity voxels in dataset CA-2 is considerably higher than in CA-1, the undersampled MIP gives a better result.

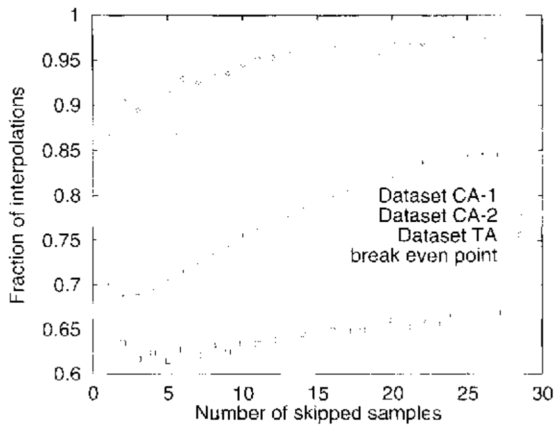

Fig. 6. Fraction of required trilinear interpolations as a function of the number of samples that are skipped during the lower-bound pass. The overhead associated with the lower-bound pass is reflected in the dashed line; when samples have a higher value than this line, the costs exceed the gains.

Our measurements suggest that the optimum sampling rate for the lower-bound pass depends on the dataset; however, sampling rates higher than 0.2 should be avoided due to the resulting overhead. We recommend using sampling rates between 0.1 and 0.02, which give a good compromise between the computational cost associated with the undersampled MIP and the potential reduction of the number of trilinear interpolations.

In our experiments, Interval Analysis using lower-bound estimation by undersampled MIP reduces the number of trilinear interpolations by 25 (CA-1) to 70% (TA) compared to standard MIP. While the latter result proves the feasibility of our approach, 3D phase contrast acquisitions are currently becoming more popular in clinical practice. The excellent signal characteristics of such acquisitions (high signal intensity with blood vessels, background noise only in non-flow areas) make them less suited for Interval Analysis.

3. Threshold-based interpolation

Even when Interval Analysis is used, the major faction of the trilinear interpolations is performed at samples that contain background noise. We can easily get rid of most background interpolations by a technique that we called Threshold-based Interpolation. With this technique, a second constraint is added to Interval Analysis: trilinear interpolation should only be performed if the sample grey-value corresponding to the maximum volume is larger than a user-specified threshold. If this constraint is not satisfied, we assume that the sample contains background noise and is not interesting.

When an uninteresting sample is encountered, one can either use nearest neighbour interpolation for the MIP calculation, or completely ignore the sample. The latter approach requires less processing, but may lead to disturbing image artifacts when a ray does not hit any interesting voxels.

Our implementation of this technique uses both alternatives. During the initial lower-bound pass, nearest neighbour interpolation is applied when samples fall below the threshold; when the actual MIP is performed, low intensity samples are ignored.

Fig. 7 summarizes the performance of the proposed method. In this figure, the grey-level threshold is normalized with respect to the cumulative histogram of the dataset; the value 1 represents the greyvalue at which 50% of the voxels has a lower greyvalue.

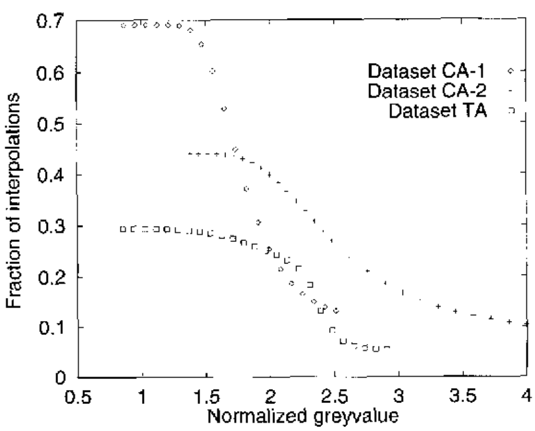

Fig. 7. Fraction of required trilinear interpolation for Threshold-based Interpolation compared with standard MIP as a function of the used grey-level threshold. The lower-bound pass was performed with a sampling rate of 0.1.

At normalized thresholds below 1.5, the reduction in the number of trilinear interpolations can be exclusively contributed to Interval Analysis. With higher thresholds, the fraction of trilinear interpolations rapidly decreases by a factor up to four. Obviously, one should not choose the threshold too high since this may result in nearest neighbour interpolation of high intensity regions, which should be avoided.

Images that result from Threshold-based Interpolation combined with Interval Analysis are almost indistinguishable from standard trilinear MIP; only the texture associated with the background noise looks somewhat different.

Ray traversal with perspective projection is expensive compared to orthographic (parallel) projection; while Threshold-based Interpolation combined with Interval Analysis is capable of reducing the number of higher-order interpolations by 80% or more, actual computational savings are only in the order of 50 to 65%.

Additional computational savings can be obtained by a technique called Ray Acceleration by Distance Coding, which is outlined in the next section.

4. Ray acceleration by distance coding

4.1. Principle

Overhead associated with ray casting can be significantly reduced by using the Ray Acceleration by Distance Coding (RADC) technique (Zuiderveld et al., 1992). This technique requires identification of interesting voxels of the input volume; a 3D distance transform is then used to create a so-called *distance volume* which contains for each voxel an approximation of the distance to the nearest interesting object.

During ray casting, these distance values are used to calculate the number of sample points that can be skipped. High distance values, typically found at off-center parts of the volume, result in rapid traversal of uninteresting regions in the volume, thus significantly reducing the number of samples to be evaluated.

Due to the large intensity differences between blood and stationary tissue in MRA acquisitions, automated segmentation of blood vessels is feasible using local filtering techniques which are tailored to recognize vessel-like structures (Vandermeulen et al., 1992). Unfortunately, local filtering methods tend to be computationally expensive and their use would only increase the total amount of time required for creating a series of MIP images. We therefore chose to use hysteresis thresholding (Gerig et al., 1993); with this technique, two different thresholds are applied to the MRA dataset. All voxels with grey-values below the lower threshold are assumed to be uninteresting, while all voxels with grey-values larger than the higher threshold are assumed to contain a major blood vessel. Smaller blood vessels with grey-levels between the two thresholds are subsequently identified by applying a region growing algorithm (using a 26-neighbour connect operator) where voxels associated with the major vessels are used as seed points.

Fig. 8 shows an example of hysteresis threshold and the resulting distance image. Suitable thresholds were obtained from the cumulative histogram of the MRA volume; the upper threshold is equal to the grey-value corresponding with the 99.95% cumulative histogram percentile, while the lower threshold is set at 97.5%.

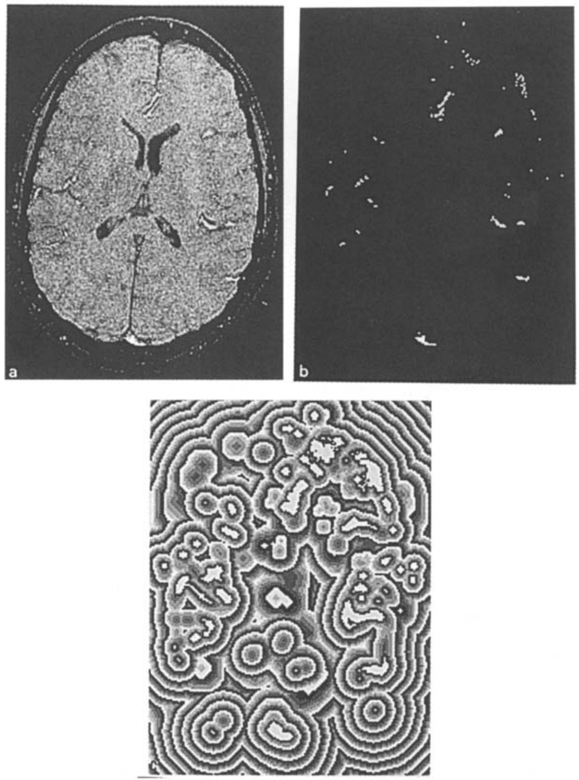

Fig. 8. Images used for RADC technique. (a) Slice of original MRA dataset. (b) Blood vessel segmentation resulting from hysteresis thresholding. (c) Calculated distance image used by the RADC technique. Grey-values in this image were assigned values such that the wave fronts can be clearly recognized.

4.2. Results of standard RADC

Results obtained by applying RADC on the three MRA datasets are summarized in Table 2; in these measurements, the preprocessing steps were not taken into account. The number of trilinear interpolations is reduced by a factor of seven to fifteen, which is the same as was achieved by threshold interpolation (see Fig. 7). Since RADC also reduces overhead associated with ray traversal, the actual improvements in execution speed are considerably higher compared to the previously discussed methods.

We evaluated the RADC method for a considerable number of additional MRA datasets and found that speed-up factors between three and ten are fairly typical. The segmentation step turns out to be important; voxels that are incorrectly labelled as interesting do not affect the quality of the MIP reconstruction, but significantly reduce the speed-up obtained by RADC. However, it is essential that all interesting blood vessels are identified; if they are missed during the segmentation step, they will not show up in the MIP image. This is illustrated in Fig. 9.

Table 2
Performance of the RADC method

Dataset	Fraction of interpolations	Improvement in execution speed
CA-1	14.2%	3.23
CA-2	14.2%	4.17
TA	5.5%	9.30

When comparing Figs. 9a and 9b, it can be seen that the major vessels are identical, while a few small vessels have not been recognized during the segmentation step and have disappeared in the MIP image.

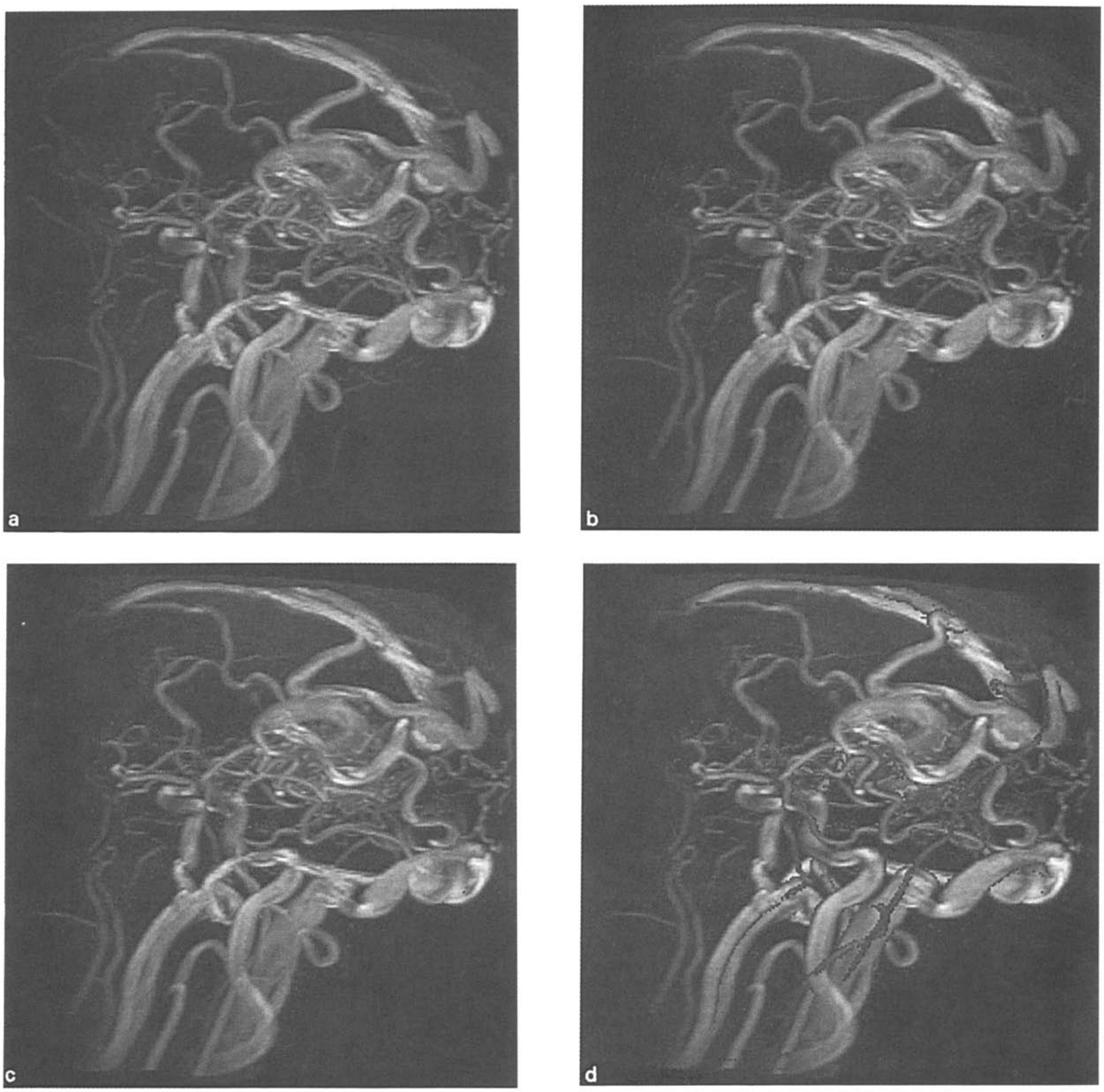

Fig. 9. Projection images calculated using a 256×256×148 MRA dataset using various techniques. (a) Standard MIP using trilinear interpolation. (b) MIP obtained using RADC. (c) MIP obtained using RADC with ray tagging. (d) MIP obtained using RADC with ray tagging combined with CVP. Enlarged sections of Figs. 9a and 9d can be found in Fig. 11.

In our experiments, we choose fairly high segmentation thresholds. While this increases the risk of missing small vessels, it significantly improves the speed of the reconstruction. However, when used clinically, users should be informed of the risk that small vessels may not be visible.

4.3. Ray tagging

A minor extension of the RADC algorithm gives additional computational savings by detecting rays that do not hit a blood vessel. The extension adds minor overhead to the ray casting step and involves keeping track of the minimum distance encountered along the ray. The key to the improved performance is the following observation: The minimum distance value that is encountered along a ray is equal to the minimum distance to the nearest interesting pixel in the output image. In other words: when the minimum distance value found along a ray is M, all rays that lie within a radius of M will not hit a blood vessel and are not interesting.

To avoid artifacts in the generated image, ray tagging is implemented using an incremental refinement technique. During a first pass, rays that track the minimum distance are cast along a sparse grid. Besides effectively labelling the background rays, the obtained grey-values for the grid points are bilinearly interpolated yielding a coarse version of the required MIP. During the subsequent pass, only non-background rays will be evaluated. Fig. 9c shows a typical result of ray tagging; while the noise in the background looks coarser, reconstruction of the vessels is not affected.

Ray tagging is especially helpful in images where only a small fraction of the image contains vessel information. The three datasets used in our experiments are extracted from larger datasets, where we omitted non-interesting volume parts. Consequently, our implementation of ray tagging only gave a small reduction (5 to 10%) in execution speed compared with RADC; if datasets are reconstructed without a bounding box, higher speed-ups can be obtained depending on the amount of high-intensity information in the dataset.

4.4. Closest vessel projection

Maximum Intensity Projection presents an inconsistent spatial perspective to the user; high intensity values obscure other features of interest and lead to irregular depth positions of the projected voxels. Depth perception can be improved by a cine-loop of MIP images reconstructed with different projection angles; however, often a stable image is desired.

There is a clear need for other algorithms for the display of vascular information; feasible approaches are the use of color (Ehricke et al., 1992) and standard volume rendering techniques using opacity blending (Vandermeulen, 1991). A different approach is Closest Vessel Projection (CVP), which limits the projection range to the lumen of the detected vessel closest to the image lane (Siebert et al., 1991). Fig. 10 explains the principle, while a typical result of CVP can be found in Fig. 11.

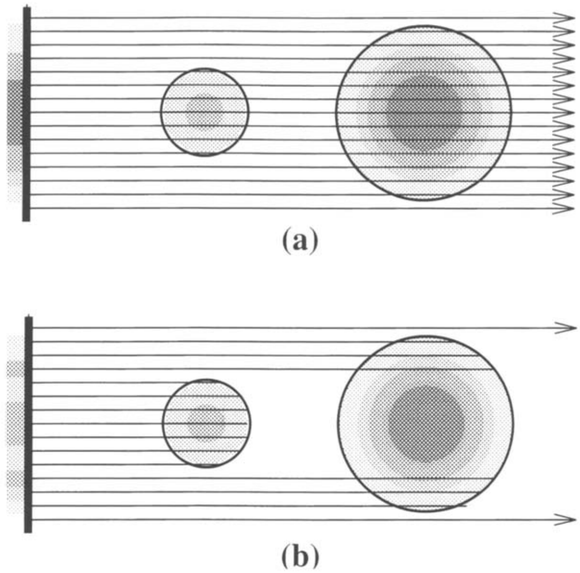

Fig. 10. Principle of Closest Vessel Projection (CVP); high image intensities are represented by dark areas. (a) With standard MIP, the smaller vessel at the left side is not visible in the output image since the larger vessel at the right side has a higher intensity. (b) With CVP, ray casting is terminated when the ray exits the first vessel it encounters. The smaller vessel is visible in the output image; it is clearly outlined because the vessel boundary is darker.

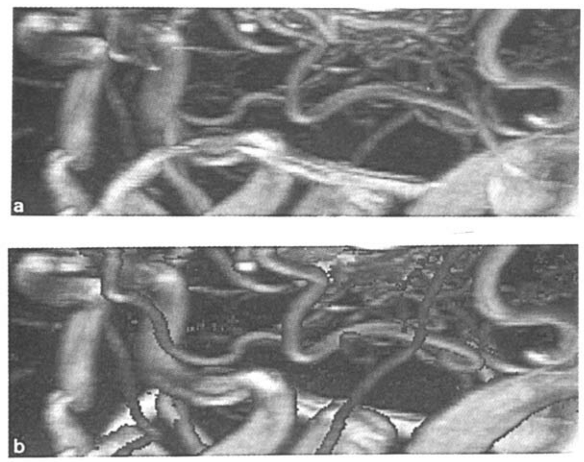

Fig. 11. Enlargements of Figs. 9a and 9d. (a) Standard MIP using trilinear interpolation. (b) MIP obtained using RADC with ray tagging and CVP. Besides its speed advantage, CVP also offers better visualization of many vessels.

Fig. 11 clearly demonstrates that CVP adds depth perception to the image. In the MIP image, a horizontal bright vessel is visible that seems to be in front of the other vessels. The CVP image clearly shows the contrary is true, while various smaller blood vessels are better delineated.

Besides an improvement in image quality, CVP also offers a small speed improvement since ray traversal can be aborted as soon as the ray leaves the first vessel it encountered. Since RADC already offers very effective ray traversal, speed improvements are only in the order of 10%.

5. Conclusions

The relative disadvantages and advantages of the proposed techniques are summarized in Table 3.

Speed improvement methods that rely on interval analysis require an additional (usually 16-bit) volume and deliver images that are identical to traditional MIP. The most successful IA method, which uses a heavily undersampled MIP to get a lower bound of the maximum value, reduces the number of trilinear interpolations to 20–50% of the original number required. Since ray casting is expensive with perspective projection, the actual savings in computational requirements is only a factor of 1.7 at best.

Threshold-based Interpolation reduces the number of trilinear interpolations to a small fraction of the original number required. It relies on a user-defined grey-level threshold; grey-values in the maximum volume that are smaller than this threshold are assumed to be background voxels, which are skipped or interpolated using nearest neighbour interpolation. The use of a threshold introduces the risk that small vessels are considered background noise; in those cases, the smaller vessels will be visualized poorly (nearest neighbour interpolation) or not at all.

The main advantage of Threshold-based Interpolation is the large reduction in trilinear interpolations compared to MIP; with perspective projection, the obtained speed-up is approximately a factor of two.

Compared with Threshold-based Interpolation, the number of trilinear interpolations obtained with Ray Acceleration by Distance Coding is approximately the same; however, the obtained speed-ups are much higher. When RADC is combined with Closest Vessel Projection, speed-up factors of 10 and higher can be obtained, while the projection images give a better depth impression than traditional MIP. While techniques as surface and volume rendering can also be applied when a high-quality segmentation is available, calculation of CVP images is less computationally expensive.

Since the distance volume (8-bit) associated with RADC requires less memory than a maximum volume (usually 16-bit), RADC is the method of choice when a fast MIP is desired. However, the quality of the obtained image depends on the reliability of the segmentation; the simple hysteresis-thresholding algorithm used in this evaluation is not suitable for very critical applications as neurosurgery where very small blood vessels should remain visible. We feel that additional work is required on obtaining a reliable segmentation of MRA data; due to the excellent and still improving image quality of modern MRA acquisitions, this should not pose a major problem.

Finally, the proposed methods are also of interest for applications that do not rely on perspective pro-

Table 3
Overview of important characteristics of the methods outlined in this paper

Technique	Fraction of interpolations	Speed improvement	Image quality
Maximum Intensity Projection (MIP)	100%	1.0	+
MIP using Interval Analysis (IA)	75–45%	1.2–1.5	+
IA with lower-bound estimation	70–20%	1.4–1.7	+
Lower-bound IA with Threshold-based Interpolation	20–5%	1.7–2.5	□/+
Ray Acceleration by Distance Coding (RADC)	20–5%	3 –10	□
Tagged RADC	20–5%	4 –13	□
Closest Vessel Projection (CVP) using tagged RADC	15–3%	4 –15	□/++

□: moderate +: good ++: excellent

jection. If parallel rays are used, the overhead associated with ray traversal is much less; consequently, the maximum speed improvements obtained by Interval Analysis and Threshold-based Interpolation will be considerably higher.

6. Acknowledgement

This research was supported in part by the Dutch ministries of Education & Science and Economic Affairs through a SPIN grant, and by the industrial companies Philips Medical Systems, KEMA and Shell.

Part of this research was funded by COVIRA (Computer Vision in Radiology), project A2003 of the AIM (Advanced Informatics in Medicine) programme of the European Community. Participants in this consortium are (in alphabetical order): Federal Institute of Technology, Zürich (CH), Foundation of Research and Technology, Crete (GR), German Cancer Research Centre, Heidelberg (D), Gregorio Maranon General Hospital, Madrid (E), IBM UK Scientific Centre, Winchester (UK), National Hospital for Neurology and Neurosurgery, London (UK), Philips Medical Systems, Best (NL) and Madrid (E); Corporate Research, Hamburg (D) (prime contractor), Royal Marsden Hospital/Institute of Cancer Research, Sutton (UK), Siemens AG, Erlangen (D) and Munich (D), University of Aachen (D), University of Hamburg (D), University of Genoa (I), University of Leuven, Neurosurgery, Radiology and Electrical Engineering (B), University of Sheffield (UK), University of Tübingen, Neuroradiology and Theoretical Astrophysics (D), and Utrecht University, Neurosurgery and Computer Vision (NL).

The phase-contrast MRA datasets used for Figs. 1b, 9, and 11 were provided by Philips Medical Systems, Best, The Netherlands.

The time-of-flight MRA acquisitions used for Figs. 1a, 2, 4, 5 and 8 were provided by Siemens Medical Systems, Erlangen, Germany.

7. References

Ehricke, H., G. Daiber, R. Sonntag, W. Straßer, M. Lochner, L. Schad and W. Lorenz (1992). Interactive 3D-graphics workstations in stereotaxy: Clinical requirements, algorithms and solutions. In: R.A. Robb, Ed., *Visualization in Biomedical Computing* 1992, SPIE 1808, 548–558.

Gerig, G., T. Koller, G. Székely, C. Brechbühler and O. Kübler (1993). Symbolic description of 3-D structures applied to cerebral vessel tree obtained from MR angiography volume data. *Proc. XIIIth Internat. Conf. on Information Processing in Medical Imaging* (*IPMI'93*), Flagstaff. 94–111.

Huston III, J. and R. Ehman (1993). Comparison of time-of-flight and phase-contrast MR neuroangiographic techniques. *RadioGraphics* 13, 5–19.

Levoy, M. (1990). Efficient ray tracing of volume data. *ACM Trans. Graphics* 9, 245–261.

Siebert, J., T. Rosenbaum and J. Pernicone (1991). Automated segmentation and presentation algorithms for 3D MR angiography. SMRM Poster 758.

Vandermeulen, D. (1991). Methods for Registration, Interpolation and Interpretation of Three-Dimensional medical image data for use in 3-D Display, 3-D Modeling and Therapy Planning. PhD thesis, Katholieke Universiteit Leuven.

Vandermeulen, D., D. Delaere, P. Suetens, H. Bosmans and G. Marchal (1992). Local filtering and global optimization methods for 3D Magnetic Resonance Angiography (MRA). In: R.A. Robb, Ed., *Visualization in Biomedical Computing* 1992, SPIE 1808, 274–288.

Zuiderveld, K. and M. Viergever (1992). Visualization of volumetric medical image data. In: P. Dewilde and J. Vandewalle, Eds., *Computer Systems and Software Engineering*. Kluwer Academic Publishers, Dordrecht, 363–385.

Zuiderveld, K., A. Koning and M. Viergever (1992). Acceleration of ray-casting using 3D distance transforms. In: R.A. Robb, Ed., *Visualization in Biomedical Computing* 1992, SPIE 1808. 324–335.